

Hawkins Experiments: Docker Quick Reference

Dockerfile Template

```
#####
# BASE IMAGE
#####
# TODO: Choose appropriate base image
# FROM <base-image>:<tag>

#####
# WORKING DIRECTORY
#####
# DO NOT MODIFY THIS SECTION
RUN mkdir /app
WORKDIR /app

#####
# SYSTEM DEPENDENCIES
#####
# TODO: Install required system dependencies
# Example for Debian-based images:
# RUN apt-get update && apt-get install -y --no-install-recommends \
#   git python3 python3-pip python3-setuptools python-is-python3 \
#   && apt-get clean && rm -rf /var/lib/apt/lists/*

#####
# REPOSITORY SETUP
#####
# TODO: Clone your repository
RUN git clone <repository-url> .
# Pin to specific commit:
# RUN git checkout <commit-sha-or-tag>
# If using submodules:
# RUN git submodule update --init --recursive

#####
# PROJECT DEPENDENCIES AND CONFIGURATION
#####
# TODO: Add project-specific dependencies

#####
# ENTRYPOINT
#####
# DO NOT MODIFY THIS SECTION
ENTRYPOINT ["/bin/bash"]
```

Common Base Images

Image	Use For
python:3.10-slim	Python projects
node:20-slim	JavaScript/TypeScript
ubuntu:22.04	Multi-language / complex setups

Basic Docker Commands

Build an image

```
docker build -t hawkins-task:v1 .
```

Run interactively

```
docker run -it --rm hawkins-task:v1
```

Run with mounted volume (to pass patch files)

```
docker run -it --rm -v $(pwd)/patch:/patch hawkins-task:v1
```

Run a script inside container

```
docker run --rm hawkins-task:v1 -c "./validation_script.sh"
```

Build without cache (for debugging)

```
docker build --no-cache -t hawkins-task:v1 .
```

Debugging

List running containers

```
docker ps
```

Shell into running container

```
docker exec -it <container_id> /bin/bash
```

View container logs

```
docker logs <container_id>
```

Inspect image layers

```
docker history hawkins-task:v1
```

Cleanup

Remove stopped containers

```
docker container prune
```

Remove unused images

```
docker image prune
```

Full cleanup (removes all unused data)

```
docker system prune -a
```